

PHYSICS-II: UNIT 2: MODEL Questions with ANSWERS

WAVE MECHANICS

Short Answer Questions.

1. What is wave particle duality?

It is well known that light exhibits the phenomena of interference, diffraction, polarization, photoelectric effect and discrete emission & absorption. The phenomena of interference, diffraction and polarization can only be explained on the basis of wave theory of light. These phenomena show that light possesses wave nature. On the other hand the phenomena of Photoelectric Effect, Compton Effect and discrete emission & absorption can only be explained on the basis of quantum theory of light, according to which light is propagated in small packets or bundles of energy. These packets are called photons or quanta and behave like corpuscles. These phenomena indicate that light possesses corpuscular (or particle) nature. Thus, the light possesses dual nature.

2. Discuss few important properties of matter waves.

- (a) Lighter particles have greater wavelength than heavier particles.
- (b) The smaller the velocity of the particle, the greater is the wavelength associated with it.
- (c) From the expression of the de Broglie wavelength, i.e. $\lambda = (h/mv)$, when $v=0$, $\lambda=\infty$, whereas if $v=\infty$, then $\lambda=0$. This shows that matter waves are generated only when particle is in motion.
- (d) Matter waves are independent of the charge. Thus, they are produced by both charged and uncharged particles. This shows that the matter waves are not electromagnetic waves; they are entirely different waves.
- (e) The velocity of matter waves is not constant. It depends on the velocity of the particle, while the velocity of electromagnetic waves is constant.
- (f) The velocity of matter waves may be greater than the velocity of light.
- (g) The wave and particle aspects of matter never appear simultaneously in the same experiment. These aspects are complementary to each other and are exhibited only in mutually exclusive experimental arrangements. In some phenomena, particle nature predominates while in other wave nature predominates.
- (h) Wave velocity of a matter wave is given in terms of group velocity as $v_{phase} = c^2/v_{group}$. Where v_{group} is equal to the particle velocity v .
- (i) Wave particle duality introduces the concept of uncertainty.

3. What was the objective of Davisson-Germer experiment? Discuss results of this experiment.

The objective of D-G experiment was to demonstrate the dual nature of moving electrons. The experiment confirmed that moving electrons display dual nature, particle as well as wave. The experimental results are discussed on the basis of different curves obtained between scattering angle and intensity of scattered beam of electrons, corresponding to different accelerating voltages V .

4. Determine the de Broglie's wavelength of electron and photons?

A light wave of frequency ν is associated with a photon of energy E , given by Plank's relation as $E = h\nu$.

According to Einstein's theory of relativity, a mass m of a particle is equivalent to an amount of energy E as $E = mc^2$

Comparing two relations of energy we get, $h\nu = mc^2$

If p is the momentum of photon, then $p = mc$

$h\nu = pc$ or $h\nu = p\lambda$ or $\lambda = h/p$. This is de Broglie wavelength of photons.

5. What is the difference between matter waves and electromagnetic waves?

(a) Matter waves are independent of the charge. Thus, they are produced for both charged and uncharged particles. This shows that the matter waves are not electromagnetic waves; they are entirely different waves.

(b) The velocity of matter waves is not constant. It depends on the velocity of the particle, while the velocity of electromagnetic waves is constant.

(c) The velocity of matter waves may be greater than the velocity of light.

6. What do you understand by phase and group velocities?

When a single wave of a definite wavelength travels in a medium, its velocity of propagation in the medium is called the wave velocity or the phase velocity. If, however, a number of waves of different wavelengths are moving with different velocities in a medium, then the observed velocity is the velocity of the wave packet (or wave group) formed by the waves. This is called the group velocity and is less than the wave velocities.

7. Show that the phase velocity of de Broglie wave is greater than velocity of light.

During the motion of matter wave, the particle and the wave associated with it are treated as complimentary to each other. Hence, there will be two different velocities; one referring to the mechanical motion of the particle represented by v and other related to the propagation of the wave represented by ω .

We know, $E = h\nu$, Also, $E = mc^2 \rightarrow h\nu = mc^2 \rightarrow \nu = mc^2/h$

The wave velocity (ω) is given by $\omega = \nu \lambda = (mc^2/h) \times (h/mv) = c^2/v \rightarrow \omega = c^2/v$

Since a particle cannot travel with velocity more than the velocity of light, thus ω will be greater than velocity of light. The satisfactory explanation of such unexpected result can be explained with the help of wave velocity and group velocity.

8. Show that de Broglie wavelength of a photon is equal to the wavelength of the radiation.

According to de Broglie, the wavelength of the photon, $\lambda(\text{photon}) = h/p$

From Einstein's relativistic energy relation, $E^2 = p^2c^2 + m_0^2c^4$, for photon $E = pc$ as $m_0 = 0$

Therefore, the wavelength of the photon = $h/p = hc/pc = hc/E = hc/h\nu = c/\nu = \lambda$ (wavelength of the radiation)

9. Explain Bohr quantization rule on the basis of de Broglie's concept of matter waves.

Bohr's quantization rule (according to which only those electron orbits are allowed as stationary orbit for which electron angular momentum is an integral multiple of $\hbar$ (or $h/2\pi$) can be explained on the basis of de Broglie concept of matter wave. In fact Bohr's

quantization conditions which were assumed arbitrary can be derived in a more natural way by the consideration of wave like property of the electron. According to Bohr, the electron in an atom revolves in circular non-radiating orbits around the nucleus. According to wave mechanical model of an atom, an electron behaves as a stationary wave which goes round the nucleus in a circular orbit. Since the electron does not radiate energy while moving in its orbit, the wave associated with it must be stationary wave in which there is no transfer of energy. According to de Broglie's concept of stationary orbit, only those orbits are allowed as stationary orbits whose circumference or perimeter is an integral multiple of wavelength associated with electron. That is $2\pi r = n\lambda$, where r is the radius of the orbit and λ the de Broglie wavelength of electron.

10. What is the physical significance of wave function?

The quantity whose variation builds up matter waves is called wave function (ψ). The value of wave function associated with a moving particle at a particular point (x, y, z) is space and time ' t ' is related to the possibility of finding the particle there at that time. The wave function enables all possible information about the particle to be derived, and being in general a complex function, does not have a direct physical meaning.

The time-dependent Schrödinger's wave equation is given by $i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m_0} \frac{\partial^2 \psi}{\partial x^2} + V\psi$

In this equation, the wave function $\psi(x, t)$ gives the behaviour of a particle at a given point x in one dimension at a given time. The magnitude of wave function is large in regions, where the probability of finding the particle is high and its magnitude is small in regions, where the probability of finding the particle is low. Thus, we can regard as a measure of the probability of finding the particle around a particular position. Now since, the probability of finding the particle at a particular region must be real and positive, but the wave function is, in general, a complex quantity; so we take $\psi\psi^* = |\psi|^2$ (where ψ^* is the complex conjugate of ψ) as a measure of the probability of finding the particle at a particular region. The quantity $|\psi|^2$ is known as the probability density.

Thus, the probability of finding the particle in a particular linear region dx is given by

$$P(x) = |\psi(x)|^2 dx$$

Similarly, the probability of finding the particle in an elementary volume dV is given by

$$P(x, y, z) = |\psi(x, y, z)|^2 dx dy dz = |\psi(x, y, z)|^2 dV$$

11. What do you understand by Eigen function and Eigen value?

An Eigen function of a linear operator, A , defined on some function in space is any non-zero function f in that space that returns from the operator exactly as it is, except for a multiplicative scaling factor. More precisely, one has $Af = \lambda f$, for some scalar, λ , the corresponding Eigen value.

12. What is the importance of Schrödinger wave equation?

Schrödinger presented his famous equation as a development of de Broglie's ideas of the wave properties. Schrödinger's equation is the fundamental equation of wave mechanics in the same sense as the Newton's second law of motion of classical mechanics. It is the differential equation for the de Broglie waves associated with particles and it describes the motion of particle. The solution of the Schrödinger wave

equation is assumed to provide a complete quantum mechanical description of the behaviour of the particle of mass m and potential energy V .

13. On the basis of Uncertainty principle, how will you explain non-existence of electrons in the nucleus?

Since the diameter of nucleus is of the order of 10^{-14} m, the maximum uncertainty for measurement of position of electron in the nucleus will be of the order of $x = 10^{-14}$ m. Using the uncertainty relation, the uncertainty in the measurement of momentum of electron is given as

$$\Delta p_x \geq h / 4\pi\Delta x = 6.63 \times 10^{-34} \text{ Js} / 4 \times 3.14 \times 10^{-14} = 0.527 \times 10^{-20} \text{ kg m/s}$$

If uncertainty in momentum of the electron is of the order of $0.527 \times 10^{-20} \text{ kg ms}^{-1}$ then the momentum of electron must be at least of the same order. Since the mass of the electron is $9.1 \times 10^{-31} \text{ kg}$, the electron in nucleus should have a velocity comparable to that of light. Using the relativistic formula for energy E of electron, we have $E^2 = p^2 c^2 + m_0^2 c^4$. Since the rest mass energy ($m_0 c^2$) is negligible (0.511 MeV) in comparison to $p^2 c^2$, the expression of energy can be given as $E^2 = p^2 c^2$ or $E = pc = [(0.527 \times 10^{-20}) \times (3 \times 10^8)] / [1.6 \times 10^{-19}] \text{ eV} = 9.88 \text{ MeV}$. The calculations show that electron can exist in the nucleus if its energy is of the order of 9.88 MeV. But we know that the electrons emitted by radioactive nuclei during γ -decay have energies of the order of 3 to 4 MeV only. Hence, electrons cannot exist in the nucleus.

14. On the basis of Uncertainty principle, how will you show that the spectral lines can never be sharp but they have a finite width?

Since an atom takes an average time period of 10^{-8} seconds to come back to its ground state from the excited state, the uncertainty in the photon energy can be given as

$$\Delta E \geq h / 4\pi\Delta t = 6.6 \times 10^{-34} / 4 \times 3.14 \times 10^{-8} = 0.525 \times 10^{-26} \text{ J}$$

Thus, the uncertainty in the frequency of light is

$$\Delta \nu \geq \frac{\Delta E}{h / 4\pi} = \frac{\Delta E \times 4\pi}{h} = \frac{0.525 \times 10^{-26} \times 4 \times 3.14}{6.6 \times 10^{-34}} = 0.999 \times 10^8 \text{ Hz} \approx 10^8 \text{ Hz}$$

The value of gives the maximum limit of accuracy with which one can determine the frequency of the radiation emitted by an atom.

15. Briefly discuss that the uncertainty principle has no meaning at macroscopic level.

Uncertainty principle has no significance in case of macroscopic objects. This is due to extremely small value of h ($6.6 \times 10^{-34} \text{ J}$). While dealing with macroscopic objects, the experimental errors involved in the measurement of position and momentum simultaneously are always such that due to our inability of measurement to this accuracy. As an example, consider a 1 gm bullet moving with a velocity 10^3 ms^{-1} . Its position might be known within an experimental accuracy of $\Delta x = 10^{-3} \text{ m}$ and its velocity to within $\Delta v = 10 \text{ ms}^{-1}$. Thus, $\Delta p = m\Delta v = (10^{-3} \text{ kg}) \times (10 \text{ ms}^{-1}) = 10^{-2} \text{ kg ms}^{-1}$

$$\Delta x \Delta p = (10^{-3} \text{ m})(10^{-2} \text{ kg ms}^{-1}) = 10^{-5} \text{ kg m}^2 \text{ s}^{-1} = 10^{-5} \text{ Js}$$

Thus value of product of uncertainty is very much greater than, $\hbar = h/2\pi = 1.005 \times 10^{-34} \text{ Js}$. Hence, the uncertainty principle has no significance for macroscopic objects.

16. What do you understand by diffraction of X-rays? Give five properties of X-rays.

When X-rays were discovered, their nature was not clear. Some considered them as high speed particles while other considered them as electromagnetic waves of short wavelength. For testing the wave nature of the X-rays, scientists tried to obtain the diffraction pattern of X-rays. But due to short wavelength of X-rays, man-made transmission gratings (ruled with 6000 lines per inch) could not produce any appreciable amount of diffraction with X-rays. Thus it was concluded that the diffraction of X-rays with man-made transmission gratings is practically impossible. Some properties of the X-rays are enumerated below:-

- (a) They are electromagnetic radiation like light. Thus they possess all the properties of electromagnetic radiations and exhibit phenomena like reflection, refraction, diffraction & interference.
- (b) Have very short range of wavelength, lying between $0.01\text{\AA}$ to $10\text{\AA}$.
- (c) Are absorbed by the material.
- (d) Are not deviated by either electric or magnetic field.
- (e) Ionize the gas through which they pass.
- (f) Can penetrate the solid material.
- (g) Affect photographic plate.
- (h) Secondary radiations arise when they strike a metal.
- (i) Have harmful effects on human tissues. Avoid continuous exposure.
- (j) Can cause fluorescence.

17. How will you show that the electron can recoil only in the onward direction while photon can scatter in any direction?

In Compton scattering, $\Delta\lambda = \lambda' - \lambda = (h/m_0c)(1 - \cos\phi)$. The expression shows that the change in wavelength is independent of the frequency of incident photon as well as of the nature of the scatterer; but depends only on the angle of scattering.

The direction of recoiled electron is $\tan \theta = \lambda \sin\phi / (\lambda' - \lambda \cos\phi)$.

Since $\lambda' > \lambda$, and $\cos\phi$ can take values from -1 to 1, the denominator will always be positive and therefore $\tan \theta$ will always be positive and less than 1. Therefore, the electron can only move in forward direction. On the other hand, $\cos\phi$ can take any value between -1 and 1 and therefore photon can scatter in any direction.

18. What is the kinetic energy of recoiled electron?

The Kinetic energy of recoiled electron is

$$KE = h \left[\frac{x(1 - \cos\phi)}{1 + (1 - \cos\phi)} \right] \text{ where } x = h\nu / m_0c^2$$

19. How will you show that only unmodified radiations are present for zero scattering angle?

In Compton scattering $\Delta\lambda = \lambda' - \lambda = (h/m_0c)(1 - \cos\phi)$. The expression shows that the change in wavelength is independent of the frequency of incident photon as well as of the nature of the scatterer; but depends only on the angle of scattering. In case of zero scattering angle, $\phi = 0 \rightarrow \cos\phi = 1 \rightarrow \Delta\lambda = \lambda' - \lambda = 0$ meaning no change in the incident wavelength. Therefore, for zero scattering, unmodified radiations are present.

Medium Answer Questions. (The answers are expected to be in 100-200 words)

1. A particle of charge q and mass m is accelerated from rest through a potential difference V . Find its de Broglie wavelength. Calculate the wavelength (λ), if particle is an electron and $V = 50$ volts. (17)

Let us consider an electron of rest mass m_0 and charge e is accelerated by the potential difference V . The kinetic energy of the electron can be given as $\frac{1}{2}m_0v^2 = eV$.

Hence, the velocity of electron will be given as $v = \sqrt{(2eV / m_0)}$.

Thus, de Broglie wavelength associated with electron can be given as

$$\lambda = \frac{h}{m_0v} = \frac{h\sqrt{m_0}}{m_0\sqrt{2eV}} = 1.23 \times 10^{-10} \text{ m} \rightarrow \lambda = 1.23 \text{ \AA}$$

2. For an electron and a proton, each having a wavelength of $1.0 \text{ \AA}$, compare their (a) momentum, (b) total energy, (c) ratio of kinetic energy. (Given that $h = 6.63 \times 10^{-34} \text{ Js}$, rest mass of electron (m_0) is $9.1 \times 10^{-31} \text{ kg}$, $c = 3.0 \times 10^8 \text{ m/s}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ joule}$). (22)

(a) de Broglie wavelength associated with a particle during its motion is given as

$$\lambda = \frac{h}{p} \text{ OR } p = \frac{h}{\lambda}$$

Now, the momentum of electron is given as $p_e = \frac{6.63 \times 10^{-34}}{1 \times 10^{-10}} = 6.63 \times 10^{-24} \text{ kg ms}^{-1}$

$$\text{For photon, } p_p = \frac{h}{\lambda} = \frac{h}{c/v} = \frac{hv}{c}$$

Thus for the same wavelength, both photon and electron have same momentum.

(b) Total energy of electron can be given as

$$\text{Total energy (E) of electron} = \text{KE} + \text{Rest mass energy} = \frac{p_e^2}{2m_e} + m_0c^2$$

$$E = \frac{(6.63 \times 10^{-24})^2}{2 \times 9.1 \times 10^{-31}} + \frac{9.1 \times 10^{-31} \times (3 \times 10^8)^2}{1.6 \times 10^{-19}} = 2.42 \times 10^{-17} + \frac{8.19 \times 10^{-14}}{1.6 \times 10^{-19}} = 0.512 \times 10^6 \text{ eV}$$

(because the first term is negligible)

Since the first term in the above equation is negligible in comparison to second one, so $E = 0.512 \times 10^6 \text{ eV} = 0.51 \text{ MeV}$

Total energy of photon (E) is given as

$$E = hv = \frac{hc}{\lambda} = p_c = 6.63 \times 10^{-24} \times 3 \times 10^8 = 1.989 \times 10^{-15} \text{ J} = \frac{1.989 \times 10^{-15}}{1.6 \times 10^{-19}} = 1.24 \times 10^4 \text{ eV} = 12.4 \text{ keV}$$

Since the rest-mass energy of photon is zero, the total energy of photon will be equal to the kinetic energy. Hence, the kinetic energy of the photon will be 12.4 keV .

(c) The kinetic energy of electron (K_e) is given as

$$K_e = 2.42 \times 10^{-17} \text{ J} = \frac{2.42 \times 10^{-17}}{1.6 \times 10^{-19}} \text{ eV} = 1.51 \times 10^2 \text{ eV} = 0.151 \text{ keV}$$

Kinetic energy of photon (K_p) is 12.4 keV. That is $K_p = 12.4 \text{ keV}$
 Now $K_e/K_p = 0.151 \text{ keV}/12.4 \text{ keV} = 1.23 \times 10^{-2}$

3. Prove that de Broglie wavelength of a particle of rest mass m_0 and charge q , accelerated by a potential difference V is given by $\lambda = \frac{h}{\sqrt{2m_0qV(1+qV/2m_0c^2)}}$. (34)

We know that de Broglie wavelength associated with a particle during its motion is given by $\lambda = \frac{h}{p}$

If a particle of charge q accelerated by a potential difference V , then kinetic energy will be qV and its rest mass energy will be m_0c^2 .

Now, the total energy of particle will be given by $E = qV + m_0c^2$

From the relativistic formula of total energy of particle, we know that $E^2 = p^2c^2 + m_0^2c^4$

From the above two equation, we can write

$$p^2c^2 + m_0^2c^4 = (qV + m_0c^2)^2 = q^2V^2 + m_0^2c^4 + 2qVm_0c^2$$

$$\text{Or } p^2 = (q^2V^2/c^2) + 2qVm_0 \text{ or } p = \sqrt{[2m_0qV\{1+qV/2m_0c^2\}]}$$

Now, using this value of p in de Broglie wave expression, we get

$$\lambda = \frac{h}{\sqrt{2m_0qV(1+qV/2m_0c^2)}}$$

4. In Davisson-Germer experiment, at what angle would the second-, and third-order diffracted beams corresponding to the strong maximum occur? At what angle would the first-order beams occur if the accelerating potential were changed from 50 volt to 65 volt? (36)

We know that the de Broglie wavelength of electron is given by $\lambda = \frac{12.28}{\sqrt{V}} \text{ \AA}$.

In D-G experiment, nickel crystal is used as grating for which inter-atomic spacing (i.e., $d=0.91 \text{ \AA}$) is $0.91 \text{ \AA}$.

Now, the wavelength of mater wave associated with electron at accelerating potential of 50 volt is given by

$$\lambda = \frac{12.28}{\sqrt{V}} = \frac{12.28}{\sqrt{50}} = 1.74 \text{ \AA}$$

From Bragg's law, we know that $2d \sin \theta = n\lambda$ or $\sin \theta = n\lambda/2d = n \times 1.74 \text{ \AA} / 2 \times 0.91 \text{ \AA} = 0.96n$

For $n=1$, $\sin \theta = 0.96$ or $\theta = \sin^{-1}(0.96) = 73.7^\circ$

For second order, $n = 2$; That is, $\sin \theta = 0.96 \times 2 = 1.92 > 1$, which is impossible because $\sin \theta$ can never exceed 1. Hence second order and third orders cannot occur.

For the first order when voltage changes from 50 to 65 volt, then

$$\sin \theta = \lambda/2d = 12.28/[1.82\sqrt{(65)}] \text{ or } \sin \theta = 0.84 \text{ or } \theta = \sin^{-1}(0.84) = 57.14^\circ$$

5. Calculate the smallest possible uncertainty in the position of an electron moving with a velocity $2 \times 10^7 \text{ m/s}$. (78)

From the principle of uncertainty, we have $\Delta x \bullet \Delta p \geq \hbar/2$ or $\Delta x \geq \hbar/4\pi\Delta p$

Given that $v = 3 \times 10^7$ m/s

$$\text{Now, } \Delta p = p = mv = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Using the value of in the equation, we get } \Delta x = \frac{0.528 \times 10^{-34} \times \sqrt{1 - \frac{v^2}{c^2}}}{m_0 v}$$

Where, $v = 3 \times 10^7$ m/s, $c = 3 \times 10^8$ m/s and $m_0 = 9 \times 10^{-31}$ kg. Using the above value in the above equation, we get

$$\Delta x = \frac{0.528 \times 10^{-34} \times \sqrt{1 - \frac{(3 \times 10^7)^2}{(3 \times 10^8)^2}}}{9 \times 10^{-31} \times 3 \times 10^7} = 0.0194 \text{ \AA}.$$

6. A particle is in motion along a line between $x=0$ and $x=a$ with zero potential energy. At point for which $x < 0$ and $x > a$, the potential energy is infinite. The wave function for the particle in the n^{th} state is given by $\psi_n = A \sin(n\pi x/a)$. Find the expression for the normalized wave function. (102)

The wave function for a particle in deep potential box is given as $\psi_n = A \sin(n\pi x/a)$.

Where n represents the n^{th} order and a is the length of line along which the particle is moving.

To find the value of a , we will use the condition of normalization as given below:-

$$\int_{-\infty}^{+\infty} |\psi_n|^2 dx = 1$$

When we use this condition, we get

$$A^2 \int_0^a \sin^2 \frac{n\pi x}{a} dx = 1 \text{ or } A^2 \int_0^a \left[1 - \cos 2\left(\frac{n\pi x}{a}\right) \right] dx = 1 \text{ or } \frac{A^2}{2} \left[1 - \frac{\sin(2n\pi x/a)}{(2n\pi/a)} \right]_0^a = 1 \text{ or } \frac{A^2}{2} \cdot a = 1$$

$$\text{or } A = \sqrt{(2/a)}$$

Using this value of A in wave function, the normalized wave function of the particle is

$$\text{given as } \psi_n = \sqrt{\frac{2}{a}} \sin(n\pi x/a)$$

7. A particle is moving in a one-dimensional box of width $30 \text{ \AA}$. Calculate the probability of finding the particle within and at intervals of $2 \text{ \AA}$ at the centre of the box when it is in its state of least energy. (109)

The wave function of a particle trapped in a deep potential box is given as

$$\psi_n = \sqrt{\frac{2}{L}} \sin(n\pi x/L)$$

In the ground state ($n=1$), particle will be at its lowest energy state. Therefore,

$$\psi_n = \sqrt{\frac{2}{L}} \sin(\pi x / L)$$

At the centre of the box, $x = L/2$. The probability of finding the particle in the unit interval at the centre of box is given as

$$|\psi_n|^2 = \left[\sqrt{\frac{2}{L}} \sin(\pi x / L) \right]^2 = \frac{2}{L} \sin^2 \pi / 2 = 2 / L$$

The probability P in the interval Δx is given as $P = |\psi_n|^2 \Delta x$ or $P = \frac{2}{L} \Delta x$

Given that $L = 30\text{\AA}$ and $\Delta x = 2\text{\AA}$, $P = (2/30) \times 2 = 0.16 = 16\%$

Long Answer Questions.

1. Discuss the dual nature of matter and waves. Describe the properties of matter waves. Also distinguish between wave & group velocity, and show that $v_g \cdot v_p = c^2$. Also, derive formulae for de Broglie wavelength for particle (of charge q and mass m) in terms of KE and its rest mass energy; accelerated through a potential difference V , and accelerated through thermal energy.
2. What are de Broglie's matter waves? Describe Davisson-Germer experiment to prove that electrons possess wave nature.
3. Explain the difference between wave velocity and group velocity in wave motion. Obtain expressions for the group velocity in dispersive and non-dispersive media.
4. Show that the phase velocity of the de Broglie wave is greater than the velocity of light, but the group velocity is equal to the velocity of particle with which waves are associated.
5. What is Uncertainty principle? Explain Heisenberg's Uncertainty principle and discuss any two applications of this principle.
6. Derive time independent and time dependent Schrödinger wave equations. Explain Ψ .
7. Derive an expression for the wave function and energy of a particle confined in one dimensional potential box using Schrödinger wave equation.
